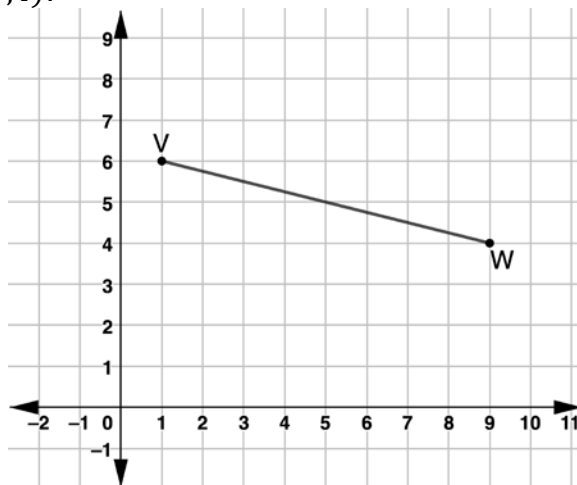


Geometry
Unit 13
Lesson 5 Practice

Name _____
Date _____
Period _____

Use the following information for questions 1-4. Point V has coordinates $(1,6)$ and point W has coordinates $(9,4)$.



1. Complete the table to calculate $\frac{1}{2}V + \frac{1}{2}W$.

	x-value	y-value
$V(1,6)$	$\frac{1}{2}(\text{---})$	$\frac{1}{2}(\text{---})$
$W(9,4)$	$\frac{1}{2}(\text{---})$	$\frac{1}{2}(\text{---})$
Total		

2. What is the midpoint of line segment VW ?
3. Complete the expression that can be used to find the point that partitions the line segment in a 1:1 ratio.

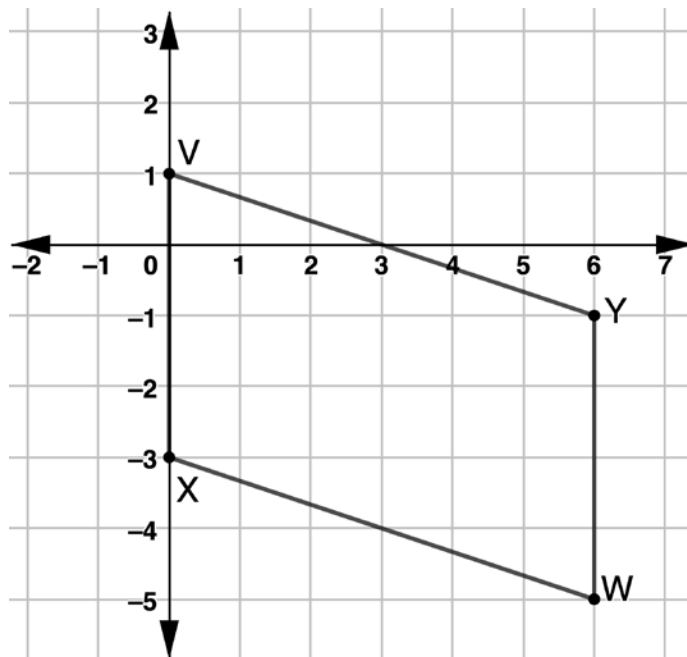
$$\frac{1}{\text{---}}V + \frac{1}{\text{---}}W$$

4. Find the point that partitions directed line segment VW in a 2:1 ratio.

Use the following information for questions 5-6. Line segment FW has endpoints at $F(-10, -10)$ and $W(6, -2)$.

5. Write an expression for the point that partitions directed line segment FW in a ratio of 3 to 5.
6. Use the expression to find the point that partitions directed line segment FW in a 3:5 ratio.

Use the following information for questions 7-10. $VXWY$ has vertices at $V(0,1)$, $X(0,-3)$, $W(6,-5)$, and $Y(6,-1)$.



7. Find the point that partitions $\overline{WX}$ in a 1:4 ratio. Label it P .
8. Find the point that partitions $\overline{WY}$ in a 1:4 ratio. Label it F .
9. Find the point that partitions $\overline{XV}$ in a 1:4 ratio. Label it R .
10. Find the point that partitions $\overline{VY}$ in a 1:4 ratio. Label it T .